Indoor Visual Place Recognition via DINOv2 Retrieval and VGGT Geometric Re-Ranking

Jia Qi (Jacky) Chen
Applied Computing
British Columbia Institute of Technology
Burnaby, BC
jchen574@my.bcit.ca

Hyeonu (Eric) Kim
Applied Computing
British Columbia Institute of Technology
Burnaby, BC
hkim505@my.bcit.ca

Henry Chen
Applied Computing
British Columbia Institute of Technology
Burnaby, BC
hchen285@my.bcit.ca

***Abstract*—Indoor visual place recognition (VPR) is the task of localizing a query image within a pre-built database of geo-tagged reference images captured in the same environment. Unlike outdoor localization, indoor scenes lack GPS signal, exhibit repetitive textures, and undergo dramatic appearance changes due to crowds and variable lighting conditions. We evaluate a two-stage retrieval pipeline on the NYC-Indoor-VPR dataset: a DINOv2-based global descriptor retrieval stage followed by geometric re-ranking with VGGT, a billion-parameter feed-forward 3-D scene reconstruction model. Three experimental conditions are compared: (A) DINOv2 zero-shot retrieval using CLS token embeddings indexed with FAISS, (B) VGGT geometric re-ranking of the top-10 DINOv2 candidates using predicted camera translation as a proximity proxy, and (C) VGGT re-ranking after triplet-loss fine-tuning on the NYC-Indoor-VPR training split. On the 1,880-query test set spanning nine indoor scenes, Condition A achieves R@1 of 10.7% at a 5 m threshold, Condition B improves marginally to 11.1%, and Condition C reaches 12.1%. All three conditions converge at R@25 = 72.2%, indicating that the primary bottleneck is top-1 precision rather than overall candidate coverage. Runtime benchmarking reveals that VGGT re-ranking adds approximately 2.15 s per query which is a 31× overhead over retrieval alone, posing a significant challenge for real-time deployment. We analyze failure modes including descriptor collapse in repetitive scenes and pedestrian occlusion, and propose future directions including scene-adaptive foreground masking and efficient learned re-ranking.**

***Index Terms*—indoor localization, visual place recognition, DINOv2, VGGT, geometric re-ranking, triplet loss fine-tuning, NYC-Indoor-VPR, FAISS**

I. Introduction

Reliable indoor localization is a fundamental capability required across a wide range of practical applications. Autonomous mobile robots deployed in warehouses and hospitals must know their location to navigate safely. Assistive devices for visually impaired users need room-level or finer localization to provide meaningful guidance. Augmented-reality systems require accurate 6-DoF camera pose estimates to overlay virtual content on the real world without drift. In all of these settings, GPS is unavailable indoors, and infrastructure-based alternatives such as WiFi fingerprinting, UWB beacons, or QR code arrays are expensive to deploy, difficult to maintain, and not universally available.

Visual place recognition (VPR) offers a compelling camera-only alternative to infrastructure-based localization. Given a query image captured by a moving agent, a VPR system retrieves the most visually similar image from a pre-built database of geo-tagged reference frames. The geographic coordinates or 6-DoF pose associated with the retrieved database image are then transferred to the query, providing a localization estimate without any additional infrastructure [1]. The accuracy of this estimate depends on how closely the retrieved image matches the query in both visual appearance and physical location.

Despite substantial progress on outdoor VPR benchmarks, indoor environments remain considerably more challenging. Several factors compound the difficulty. First, many indoor spaces are visually repetitive: library stacks, subway corridors, and open-plan office floors all contain similar colors, textures, and structural elements that make different locations look nearly identical. Second, indoor scenes are dynamic: crowds of people cause heavy occlusion between sessions, and changing artificial lighting alters the appearance of surfaces significantly. Third, indoor trajectories often involve elevators, stairs, and sharp turns, producing large viewpoint changes between nearby locations. Together, these factors mean that even strong vision-transformer-based descriptors, which have driven impressive progress on outdoor benchmarks, struggle to achieve R@1 above 20% within a 5 m tolerance on recent indoor benchmarks such as NYC-Indoor-VPR [2].

A natural response to low single-image descriptor performance is to introduce a second stage that verifies geometric consistency between a query and its candidate database images. Classical two-stage pipelines retrieve a shortlist using a fast global descriptor and then re-rank using local feature matching and geometric verification [8]. This approach has proven effective outdoors, but its application indoors is less studied, partly because keypoint detectors trained on outdoor imagery perform poorly on indoor textures, and partly because indoor datasets with dense ground-truth pose labels have only recently become available.

In this work, we investigate whether VGGT [4], a recently

proposed 1.25-billion parameter transformer that jointly predicts camera poses and dense 3-D structure from image pairs in a single feed-forward pass, can serve as an effective geometric re-ranker for indoor VPR. VGGT is attractive for this role because it does not require explicit keypoint detection or matching and generalizes across scene types without per-scene optimization. We evaluate it on top of DINOv2 [3] global retrieval and additionally ask whether fine-tuning VGGT on the target indoor domain can close the remaining gap.

This paper makes the following contributions:

- We establish a strong DINOv2 zero-shot retrieval baseline on the NYC-Indoor-VPR benchmark using CLS token embeddings, FAISS exact search, and a comprehensive evaluation across three distance thresholds and four K values.
- We propose and implement a VGGT-based geometric re-ranking stage that uses predicted inter-image camera translation as a proximity score, and define a normalized score fusion strategy to combine retrieval and geometric signals.
- We perform domain adaptation of VGGT via triplet margin loss fine-tuning on the NYC-Indoor-VPR training split, unfreezing the top four transformer blocks and all normalization layers, and evaluate the impact on test-set recall.
- We conduct a thorough runtime analysis of the full pipeline, identifying VGGT re-ranking as the dominant latency bottleneck and estimating the cost of each stage at query time.
- We provide a per-scene breakdown of recall performance, identifying structural factors that explain the large variance across scenes, and propose two concrete directions for future improvement.

The remainder of the paper is organized as follows. Section II reviews related work in global descriptor retrieval, geometric re-ranking, and 3-D scene understanding. Section III describes the dataset, the full pipeline architecture, and the fine-tuning procedure in detail. Section IV presents quantitative results, visualizations, and runtime measurements. Section V analyses failure modes, limitations, and future directions. Section VI summarizes the key findings.

II. Related Work and Background

A. Global Descriptor Retrieval for VPR

The dominant paradigm in VPR is to represent each image as a compact global descriptor and perform nearest-neighbor search over a database of such descriptors at query time. Early work aggregated local hand-crafted features—SIFT, ORB, BRIEF—into bag-of-words histograms [5], which could be searched efficiently using inverted index structures. However, bag-of-words representations discard spatial layout and are sensitive to viewpoint and lighting changes, limiting their performance in challenging conditions.

NetVLAD [1] was a turning point: it introduced a differentiable VLAD pooling layer that could be trained end-to-end using weakly supervised triplet loss on geo-tagged street imagery. By learning to pool convolutional features into a fixed-size descriptor, NetVLAD substantially outperformed hand-crafted baselines on standard outdoor benchmarks such as Pittsburgh and Tokyo 24/7. Follow-up work such as CosPlace [6] improved descriptor compactness and training efficiency by reformulating the learning objective as a classification problem over geographic cells, while MixVPR [7] proposed a simple feature mixing architecture that achieves competitive performance with reduced model complexity.

More recently, self-supervised vision transformers have emerged as powerful general-purpose visual encoders. DINOv2 [3] pre-trains a ViT-B/14 architecture using a combination of self-distillation and masked image modeling on a curated dataset of 142 million images. The resulting CLS tokens capture rich semantic and structural information and transfer effectively to downstream tasks without fine-tuning. Several studies have shown that DINOv2 features achieve strong zero-shot VPR performance, in some cases matching or exceeding methods trained specifically for place recognition. However, most evaluations focus on outdoor benchmarks; indoor performance under the challenges described in Section I remains undercharacterized.

B. Two-Stage Hierarchical Localization

A well-established strategy for improving VPR precision is to split localization into a coarse retrieval stage and a fine re-ranking stage [8]. The retrieval stage rapidly identifies a shortlist of K candidates using a fast global descriptor and approximate nearest-neighbor search. The re-ranking stage then applies a more computationally expensive but geometrically precise verification step to reorder the shortlist.

Classical re-ranking methods apply RANSAC-based geometric verification using local feature correspondences established by SIFT or ORB matching. This is effective when the scene contains sufficient texture and the viewpoint change between query and candidate is moderate, but breaks down in texture-poor or highly dynamic indoor scenes.

SuperGlue [9] replaced handcrafted matching with an attention-based graph neural network that jointly predicts correspondences and their confidence scores, significantly improving robustness to viewpoint change and partial occlusion. LightGlue subsequently made learned matching faster and more parameter-efficient. These methods, however, still depend on a keypoint detector (typically SuperPoint) to extract candidate correspondences, and keypoint detection quality degrades in featureless or occluded regions common in indoor images.

Our work differs from these approaches by using VGGT as the re-ranking signal. Rather than establishing explicit correspondences, VGGT directly predicts relative camera pose from image pairs in a feed-forward pass, bypassing keypoint detection entirely and enabling re-ranking of panoramic images for which standard keypoint detectors are not well calibrated.

C. VGGT: Visual Geometry Grounded Transformer

VGGT [4] is a 1.25-billion parameter transformer model that takes an unordered set of images as input and jointly predicts, for each image, dense depth maps, per-pixel 3-D world point clouds, and camera extrinsic and intrinsic parameters. Unlike structure-from-motion or NeRF-based methods [10] that require iterative per-scene optimization, VGGT performs inference in a single feed-forward pass and generalizes across diverse scene types without retraining. The model architecture consists of a shared image encoder followed by a cross-image aggregator that allows each image's features to attend to all other images in the input set, enabling consistent multi-view geometry prediction.

In our pipeline, we exploit VGGT in two distinct ways. In Conditions B and C, we use VGGT as a two-image geometric scorer: given a query image and a candidate database image, VGGT predicts camera translation vectors for both, and the magnitude of the predicted relative displacement serves as a geometric proximity score. In Condition C, we additionally adapt VGGT's internal aggregator as a descriptor extractor for triplet-loss fine-tuning, using mean-pooled aggregator tokens as global image embeddings.

D. NYC-Indoor-VPR Dataset

NYC-Indoor-VPR [2] is, to our knowledge, the largest publicly available dataset for indoor visual place recognition. It consists of 360-degree panoramic images captured across 20 diverse indoor environments in New York City using a lidar-inertial SLAM system that provides ground-truth metric (x, y) positions for every frame. The environments span a wide range of scene types: university library floors with densely packed bookshelves, a busy subway station, a supermarket, open office floors at multiple elevations, and museum gallery spaces. This diversity makes NYC-Indoor-VPR a demanding benchmark that exercises a system's ability to generalize across very different visual domains.

The dataset provides standardized train, validation, and test splits, and each split is further divided into database images (whose positions are known) and query images (whose positions are to be estimated). Ground-truth positions enable topometric evaluation at configurable distance thresholds, which is more informative than binary scene-level accuracy for deployment in large indoor spaces. The test split used in our evaluation contains 1,939 database images and 1,880 query images across nine scenes, making it a representative and practically meaningful evaluation set.

III. Methodology and Architecture

A. Dataset Preparation and Splits

The full NYC-Indoor-VPR dataset contains 34,777 images across 20 scenes, with a combined database and query count split as follows: 15,601 training database images, 7,630 training query images, 3,941 validation database images, 3,786 validation query images, 1,939 test database images, and 1,880 test query images. The number of images per scene varies substantially, from 72 images in The Met 3rd region to 11,890 in the World Trade Center.

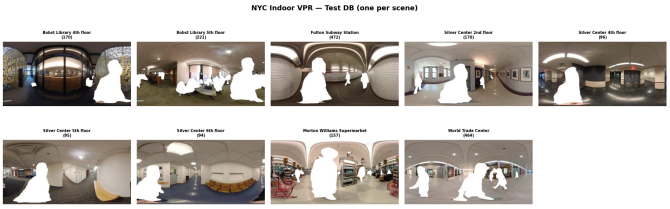

Fig. 1. Representative database images from the nine test scenes in NYC-Indoor-VPR. The diversity of visual environments—library floors, subway stations, supermarkets, office buildings—highlights the generalization challenge faced by any single descriptor model.

Each image filename encodes its ground-truth (x, y) metric position, which we parse and store in a metadata DataFrame alongside the scene label, split, and category (database or query). All file paths are verified at load time. A 360-degree panoramic JPEG is associated with each record.

For all final quantitative evaluations, we use only the test split (nine scenes). Training and validation splits are used exclusively for fine-tuning (Condition C) and hyperparameter selection respectively. The use of separate database and query sets within each split prevents any overlap between the images used for indexing and those used for querying.

B. Stage 1: DINOv2 Global Retrieval (Condition A)

The first stage of our pipeline extracts a compact global descriptor from each image and uses approximate nearest-neighbor search to identify candidate matches. We use the `facebook/dinov2-base` checkpoint (ViT-B/14, 86 million parameters) loaded from HuggingFace Transformers without any task-specific modification. Images are pre-processed using DINOv2's standard `BitImageProcessor`, which resizes and normalizes each image to the model's expected input distribution.

Each image is passed through the ViT encoder, and the CLS token from the final transformer layer is extracted as a 768-dimensional descriptor. This token aggregates global context from all patch tokens via the self-attention mechanism and has been shown in prior work to encode scene-level semantics effectively. The descriptor is L2-normalized before storage and retrieval, so inner-product similarity is equivalent to cosine similarity.

All 1,939 test-database descriptors are indexed offline in a FAISS `IndexFlatIP` structure, which performs exact inner-product search without approximation. While approximate search structures such as HNSW or IVF would offer lower query latency at larger database scales, the test database of 1,939 images is small enough that exact search completes in under 5 ms and introduces no approximation error.

At query time, the 1,880 test-query images are embedded in batches of 64 on an NVIDIA A100-SXM4-80 GB GPU, achieving approximately 2.3 batches per second. For each

query, the top-25 most similar database descriptors are retrieved. The top-25 shortlist is passed to Stage 2 for re-ranking (top-10) or returned directly as the final ranking for Condition A.

C. Stage 2: VGGT Geometric Re-ranking (Condition B)

For each query, the top-10 candidates from the DINOv2 shortlist are re-ranked using a geometric score derived from VGGT. The key insight is that if a query image and a database image depict the same physical location from a similar viewpoint, a model that predicts relative camera pose from the image pair should estimate a small inter-camera displacement. Conversely, a large predicted displacement suggests the two images were captured far apart, even if their global descriptors happen to be similar.

Concretely, VGGT is invoked with a two-image input tensor of shape $(2, 3, H, W)$ containing the query and a candidate image side by side. The model predicts camera extrinsic matrices for both images simultaneously. We extract the translation vectors $\mathbf{t}^q$ and $\mathbf{t}^d$ from the predicted extrinsics and compute the geometric score as the negated Euclidean distance:

$$s_{\text{geo}}(q, d) = -\|\mathbf{t}^q - \mathbf{t}^d\|_2 \tag{1}$$

A more negative s_{geo} (larger predicted displacement) indicates a worse geometric match. Since retrieval scores and geometric scores have different scales and distributions, we apply min-max normalization over the candidate pool before combining them:

$$\hat{s} = \frac{s - \min(s)}{\max(s) - \min(s) + \epsilon} \tag{2}$$

The final re-ranking score for each candidate is a weighted sum:

$$s_{\text{final}}(q, d) = \alpha \cdot \hat{s}_{\text{ret}}(q, d) + (1 - \alpha) \cdot \hat{s}_{\text{geo}}(q, d) \tag{3}$$

where $\alpha = 0.5$ for Condition B (equal weighting of retrieval and geometric signals). Candidates 11 through 25 from the DINOv2 shortlist are not re-ranked and are appended in their original order after the re-ranked top-10.

The ten sequential VGGT calls required to re-rank each query's top-10 candidates are the primary source of latency in the pipeline. Each call processes a single image pair through the full 1.25B parameter model, requiring approximately 215 ms on the A100 GPU, for a total re-ranking time of roughly 2.15 s per query.

D. Stage 3: Fine-tuned VGGT Re-ranking (Condition C)

Our third condition investigates whether fine-tuning VGGT on the NYC-Indoor-VPR training domain can improve the quality of its geometric re-ranking signal. We adapt VGGT's internal cross-image aggregator as a global image descriptor extractor: the output tokens of the final aggregator layer are mean-pooled across the sequence dimension to produce a single embedding vector per image, which is then L2-normalized.

Triplet dataset construction. We construct a triplet dataset from the 15,601 training database images. For each anchor image a, positive examples p are sampled from database images in the same scene within 3 m, and negatives n are sampled from database images in the same scene beyond 10 m. Up to three triplets are formed per anchor, yielding 46,661 triplets in total. The positive threshold of 3 m and negative threshold of 10 m are chosen to create a clear margin that encourages the model to learn proximity-sensitive embeddings without training on hard negatives that are geometrically ambiguous.

Training configuration. To preserve VGGT's general-purpose geometric reasoning ability while adapting it to the indoor distribution, we selectively unfreeze a subset of the model's parameters. Specifically, we unfreeze the last four transformer blocks (layers 20–23 of the 24-layer backbone) and all parameters whose names contain `head`, `norm`, or `output`. This yields 498 million trainable parameters out of 1.25 billion total. Freezing the lower layers retains the low-level visual features learned during pre-training while allowing the upper layers to adapt to the statistical properties of indoor panoramic imagery.

Training uses the AdamW optimizer with a learning rate of 5×10^{-6} and weight decay of 10^{-4}. Gradients are accumulated over two mini-batches (effective batch size 32) to stabilize updates with the large model. Mixed precision (bfloat16) is used throughout, and gradient norms are clipped at 1.0 to prevent instability in the early epochs. The triplet margin loss is:

$$\mathcal{L} = \max(0,\ \|\mathbf{e}_a - \mathbf{e}_p\|_2 - \|\mathbf{e}_a - \mathbf{e}_n\|_2 + m) \tag{4}$$

where $\mathbf{e}_a$, $\mathbf{e}_p$, $\mathbf{e}_n$ are the normalized embeddings of the anchor, positive, and negative images respectively, and $m = 0.3$ is the margin.

To accelerate training, the anchor, positive, and negative images in each triplet are stacked into a single batch of shape $(3B, 3, H, W)$ and passed through the model in one forward pass, after which the output is split back into the three components. This avoids three separate forward passes per triplet and improves GPU utilization significantly. Training runs for three epochs over the full triplet dataset, with each epoch processing 2,917 batches. Total training time is approximately 13.7 hours on the A100 GPU.

At evaluation time, Condition C uses the same VGGT re-ranking pipeline as Condition B but with the fine-tuned model weights and a slightly upweighted geometric signal ($\alpha = 0.6$, giving 60% weight to the retrieval score and 40% to the geometric score).

IV. Experiments and Results

A. Experimental Setup

All experiments are conducted on Google Colab Pro+ with an NVIDIA A100-SXM4-80 GB GPU (80 GB HBM2e), PyTorch 2.10.0+cu128, CUDA 13.0, and bfloat16 mixed precision for all model inference. The random seed is fixed to

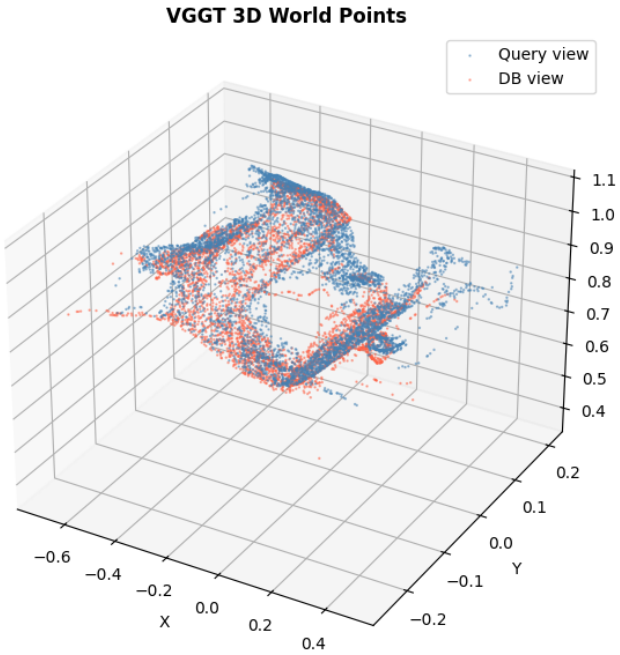

Fig. 2. 3D world points predicted by VGGT for a query-database image pair. Blue points are from the query view, red from the database view. Overlapping point clouds indicate geometric consistency between the two images, which we use as a proximity signal for re-ranking.

42 across NumPy, Python's `random` module, and PyTorch to ensure reproducibility.

Metrics. We report Recall@K (R@K) for $K \in \{1, 5, 10, 25\}$ at distance thresholds $d \in \{3, 5, 10\}$ m. A query is considered correctly localized at rank K and threshold d if at least one of the top-K retrieved database images (a) belongs to the same scene as the query and (b) lies within d m of the query's ground-truth position. This topometric criterion is more informative than binary scene-level accuracy for large indoor spaces where the same scene label covers tens of metres of traversable area.

All evaluations are run on the test split only. Validation recall is computed on a 100-image sample during fine-tuning for monitoring purposes, but hyperparameter selection is kept minimal to avoid test-set contamination.

B. Overall Recall Results

Table I reports R@K across all three conditions and all three distance thresholds. At the primary threshold of 5 m, the DINOv2 zero-shot baseline (Condition A) achieves R@1 = 10.7%, R@5 = 38.1%, R@10 = 52.3%, and R@25 = 72.2%. VGGT geometric re-ranking (Condition B) raises R@1 slightly to 11.1% while leaving R@5 and above largely unchanged. Fine-tuned VGGT re-ranking (Condition C) achieves the best R@1 = 12.1%, a 1.4 percentage point gain over the zero-shot baseline.

A consistent pattern across all conditions is that R@25 is substantially higher than R@1. This gap—roughly 60 percentage points at 5 m—reveals that the correct database image is

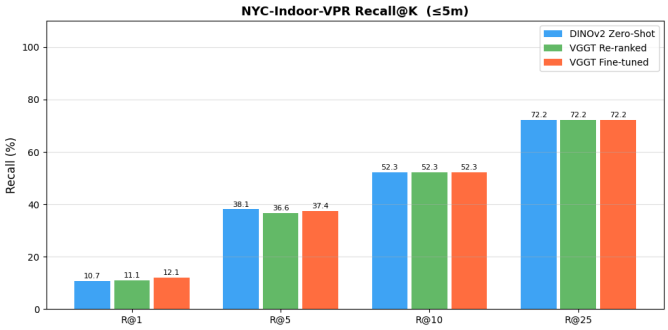

Fig. 3. Recall@K at the 5 m threshold for all three conditions. The gap between R@1 and R@25 highlights that correct candidates are retrieved but rarely ranked first.

TABLE I
RECALL@K (%) ON THE NYC-INDOOR-VPR TEST SET (1,880 QUERIES, 9 SCENES). **BOLD**: BEST RESULT PER COLUMN AT 5 M THRESHOLD.

Condition	R@1	R@5	R@10	R@25
Distance threshold = 3 m				
A: DINOv2 Zero-Shot	7.3	26.9	38.2	54.0
B: VGGT Re-ranked	7.3	25.6	38.2	54.0
C: VGGT Fine-tuned	8.1	26.5	38.2	54.0
Distance threshold = 5 m				
A: DINOv2 Zero-Shot	10.7	**38.1**	52.3	72.2
B: VGGT Re-ranked	11.1	36.6	52.3	72.2
C: VGGT Fine-tuned	**12.1**	37.4	52.3	**72.2**
Distance threshold = 10 m				
A: DINOv2 Zero-Shot	17.5	54.8	72.7	91.3
B: VGGT Re-ranked	18.0	55.2	72.7	91.3
C: VGGT Fine-tuned	**18.8**	**55.2**	72.7	91.3

almost always present in the top-25 shortlist but rarely ranked first. Re-ranking successfully promotes the correct answer to rank 1 in some additional cases (Conditions B and C both improve over A at R@1), but the majority of re-ranking decisions do not change the top-1 outcome. At the 10 m threshold, Condition C achieves R@1 = 18.8% and R@25 = 91.3%, suggesting that at a more relaxed precision requirement the pipeline performs considerably better.

C. Per-Scene Analysis

Fig. 4 shows the per-scene R@1 at 5 m for all three conditions. Performance varies dramatically across scenes, spanning from near 0% at the World Trade Center and Fulton Subway Station to approximately 30–35% at Silver Center 4th Floor. This variance reflects genuine differences in scene structure rather than dataset artifacts: scenes with distinct spatial layout and varied visual content (Silver Center floors, Bobst Library floors) are substantially easier for the retrieval pipeline than scenes with large spatial extent and homogeneous appearance (World Trade Center, subway corridors).

The fine-tuned model (Condition C) achieves its largest gains over the zero-shot baseline at Silver Center 5th and 6th floors and at Bobst Library 5th floor. This suggests that fine-tuning helps most in scenes where the structural cues are

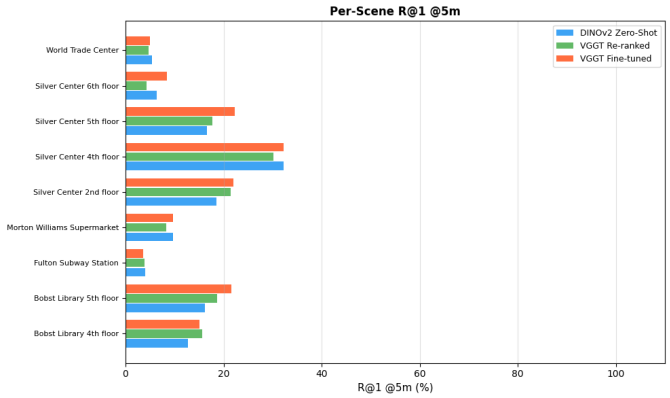

Fig. 4. Per-scene R@1 at 5 m for all three conditions. Scenes are sorted by name. The World Trade Center and Fulton Subway Station are the most challenging, while Silver Center floors achieve the highest recall.

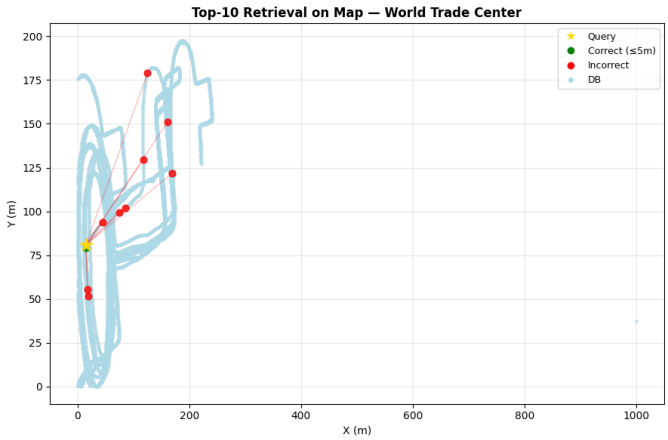

Fig. 5. Top-10 retrieval results plotted on the topometric map of the World Trade Center scene. The gold star marks the query location. Green dots are correct retrievals (within 5 m); red dots are incorrect. The cluster of red dots spread across the map illustrates the descriptor collapse problem in large repetitive environments.

present but require domain adaptation to exploit, rather than in scenes where the visual content is simply too repetitive for any descriptor to discriminate.

Notably, Condition B (VGGT re-ranking without fine-tuning) occasionally performs worse than the DINOv2 baseline at R@5, most visibly at the 5 m threshold. This happens because re-ranking can demote correct candidates from ranks 2–5 to lower positions, reducing R@5 even when R@1 improves. The fine-tuned model (Condition C) partially recovers R@5 relative to Condition B by producing more reliable geometric scores, but still does not consistently match the baseline at R@5.

D. Runtime Analysis

Table II presents per-query latency statistics measured over 30 randomly sampled test queries. DINOv2 descriptor extraction for a single query and exact FAISS search over 1,939 database vectors completes in a mean of 69.7 ms (median 69.5 ms, P95 72.0 ms), which is fast enough for interactive use. VGGT re-ranking of the top-10 candidates adds a mean of 2,152.5 ms (P95 2,170.8 ms), bringing the total pipeline latency to 2,222.2 ms per query.

TABLE II
PER-QUERY PIPELINE LATENCY (MS), MEASURED OVER 30 TEST QUERIES ON AN NVIDIA A100-SXM4-80 GB GPU.

Stage	Mean	Median	P95
DINOv2 embed + FAISS	69.7 ms	69.5 ms	72.0 ms
VGGT re-ranking (top-10)	2152.5 ms	—	2170.8 ms
Total	2222.2 ms	—	—

The VGGT re-ranking overhead is a 31× multiplier relative to retrieval alone. At its current latency, the pipeline operates at roughly 0.45 queries per second, which falls well below the real-time requirements of most robotics or navigation applications (typically ≥5 Hz). The latency is dominated by the ten sequential VGGT forward passes required to score each query's top-10 candidates. Each individual two-image VGGT call takes approximately 215 ms; the remainder is overhead from Python-level looping, CUDA synchronization, and image pre-processing.

E. Fine-tuning Dynamics

The triplet loss decreases monotonically from 0.246 (epoch 1) to 0.224 (epoch 2) to 0.212 (epoch 3), confirming that the model is learning to separate positive and negative pairs in the NYC-Indoor-VPR embedding space. Validation R@1 at 5 m, measured on a 100-image sample at the end of each epoch, improves from 0.0% (epoch 1) to 2.0% (epoch 2) to 4.0% (epoch 3).

The training loss curve is smooth and decreasing, with no signs of instability or overfitting within the three-epoch window. However, validation R@1 remains low in absolute terms, suggesting that three epochs of fine-tuning is insufficient to fully adapt the model to the indoor domain. The gap between training-split triplet separability (evidenced by a decreasing loss) and test-split R@1 implies that the fine-tuned descriptor generalizes imperfectly to the nine test scenes, which include environments not well represented in the training triplets.

V. DISCUSSION AND ANALYSIS

A. Why Re-ranking Gains Are Small

The most important observation from our results is that VGGT re-ranking produces only modest improvements in R@1 and, in some cases, degrades R@5. Understanding why requires examining both the properties of the retrieval stage and the limitations of the geometric scoring approach.

The DINOv2 retrieval stage already achieves R@25 = 72.2% at 5 m, meaning the correct answer is in the top-25 for the large majority of queries. The bottleneck is therefore not candidate recall but ranking precision: the pipeline must promote the correct candidate from somewhere in the top-25 to rank 1. This is a harder problem than simply filtering out irrelevant candidates, because the correct candidate may

be visually similar to other top-ranked candidates that are physically nearby.

The VGGT geometric score is fundamentally limited by the fact that VGGT was pre-trained on outdoor multi-view datasets with natural lighting and perspective cameras. When applied to indoor 360-degree panoramic images, VGGT has no prior knowledge of indoor metric scale, camera height, or the characteristic depth ranges encountered in indoor scenes. As a result, the predicted translation magnitudes reflect VGGT's learned priors about camera motion rather than the actual physical distances between indoor locations. This makes s_{geo} a noisy proximity indicator that can rank geometrically distant candidates above nearby ones.

A further limitation is that VGGT processes each query-candidate pair independently. Without a shared coordinate frame across all candidates, the geometric scores for different candidates are not mutually consistent. A true geometric re-ranker would ideally reason about all candidates jointly—for example, by enforcing that their predicted poses are consistent with a single coherent scene model—but VGGT's pair-wise inference does not support this.

B. Failure Mode Analysis

Repetitive scenes. The World Trade Center and Fulton Subway Station are the most challenging scenes in the test set, with near-zero R@1 at 5 m. Both environments share the characteristic of large spatial extent combined with visually homogeneous appearance: the World Trade Center spans over 1 km in the x-dimension, and its corridors and lobbies exhibit similar reflective marble surfaces, recessed overhead lighting, and open atrium geometries throughout. The Fulton Subway Station has a similar structure: long platforms with uniform tiling and consistent artificial lighting. In these environments, DINOv2's CLS token, which pools global semantic information from all image patches, produces very similar descriptors for locations that are tens of metres apart, making the retrieval problem geometrically ill-posed. VGGT re-ranking cannot help when the retrieval shortlist is dominated by incorrect candidates that are visually indistinguishable from the correct one.

Masked pedestrian regions. The NYC-Indoor-VPR dataset was captured in active public spaces and all pedestrians have been anonymized as uniform white silhouettes prior to release. While this addresses privacy concerns, the white silhouette regions still occupy large fractions of the image area in many frames and introduce a different kind of appearance inconsistency. Because database and query images were captured at different times, the number, size, and position of white silhouette regions varies significantly between a query and its correct database match. DINOv2's CLS token aggregates information from all image patches, including those dominated by these silhouette regions. A query image with heavy silhouette coverage may therefore produce a descriptor that is closer to other heavily-occluded database images regardless of their physical location, rather than to the geographically correct match with a clear field of view. This cross-session appearance variation remains a challenge in VPR [1] even when the content causing it has been anonymized rather than removed entirely.

VGGT score degradation at R@5. As noted in Section IV, Condition B occasionally degrades R@5 relative to Condition A. This occurs when the correct candidate sits at rank 2–5 in the DINOv2 shortlist and receives a poor geometric score because VGGT's pose prediction for the query-correct pair is unexpectedly large. In such cases, re-ranking demotes the correct candidate while promoting an incorrect one with a fortuitously small predicted displacement. This suggests that the geometric score is not reliable enough to override strong retrieval evidence for individual candidates, particularly given that VGGT was pre-trained on outdoor scenes and does not have a reliable sense of indoor metric scale.

C. Limitations of the Current Approach

Several aspects of our implementation constrain the achievable performance. First, the triplet dataset for fine-tuning is constructed entirely from training database images; query images are never seen during training. Because database and query images are captured at different times under different crowd and lighting conditions, there may be a domain shift between training descriptors and the query descriptors encountered at test time. Including query images from the training split in the fine-tuning data would likely improve generalization.

Second, the fine-tuning schedule of three epochs with only the top four transformer blocks unfrozen is conservative. A more aggressive fine-tuning strategy—longer training with learning rate warm-up, full-model unfreezing, or data augmentation simulating indoor lighting changes—might yield greater improvements, but was impractical within the resource constraints of this project.

Third, the evaluation is conducted on a single GPU in a sequential fashion, which artificially inflates latency for the VGGT re-ranking stage. A production implementation would batch candidate pairs and process them in parallel, potentially reducing re-ranking time by an order of magnitude.

D. Directions for Future Work

Three concrete research directions emerge from our analysis.

Fine-tuning DINOv2 on indoor data. The retrieval stage is where most of the performance gap originates — DINOv2 was used completely out of the box with no indoor-specific adaptation. Fine-tuning DINOv2 directly on NYC-Indoor-VPR training images using a contrastive or triplet objective would encourage the model to produce descriptors that are sensitive to the metric distances between indoor locations rather than purely semantic similarity. Since R@25 already reaches 72.2%, even a modest improvement in descriptor discriminability at short distances could translate directly into higher R@1, which is the primary bottleneck.

Efficient learned re-ranking. The 2.15 s VGGT re-ranking overhead makes the current pipeline impractical for real-time

use. A promising direction is to run VGGT offline over the entire database once to extract per-image feature representations, which would then be stored and compared against query features using a small lightweight network at query time. This approach amortizes the cost of VGGT inference over the database rather than repeating it for every query, and reduces query-time re-ranking from ten sequential billion-parameter model calls to a series of lightweight network evaluations, potentially achieving geometric re-ranking at tens-of-milliseconds latency.

VI. Conclusion

We presented and evaluated a two-stage indoor visual place recognition pipeline on the NYC-Indoor-VPR benchmark. The pipeline combines DINOv2 zero-shot global descriptor retrieval with VGGT geometric re-ranking, and we further investigated the effect of fine-tuning VGGT on the target indoor domain using triplet margin loss. Our main findings are as follows.

DINOv2 zero-shot retrieval establishes a competitive baseline, achieving R@25 = 72.2% at 5 m across nine challenging indoor scenes. VGGT geometric re-ranking provides a consistent but modest improvement in R@1 (+0.4 percentage points at 5 m) while occasionally reducing R@5 due to unreliable geometric scores. Fine-tuning VGGT on the indoor domain yields the best overall performance (R@1 = 12.1% at 5 m, +1.4 pp over baseline) and partially mitigates the R@5 degradation introduced by re-ranking. All three conditions converge at R@25, confirming that improving top-1 precision—not candidate recall—is the key open challenge.

Runtime analysis reveals that VGGT re-ranking is 31× slower than retrieval alone, with a total pipeline latency of 2.22 s per query. This makes the current system unsuitable for real-time deployment and motivates future work on efficient re-ranking strategies.

The per-scene breakdown highlights two dominant failure modes: descriptor collapse in visually repetitive environments (World Trade Center, Fulton Subway Station) and cross-session appearance change due to pedestrian occlusion. Addressing these failure modes through foreground-aware descriptor learning and domain-adaptive training represents the most promising path toward practical indoor localization performance.

Author Contributions

Hyeonu (Eric) Kim led the design and implementation of the VGGT geometric re-ranking pipeline (Condition B), including the geometric scoring function, normalized score fusion, and the live demo script. He was the primary author of the Methodology and Architecture and Discussion and Analysis sections, and contributed to the paper throughout.

Jacky Chen led the implementation of the DINOv2 retrieval baseline (Condition A), including descriptor extraction, FAISS indexing, and the full evaluation harness for topometric Recall@K. He was the primary author of the Related Work and Background and Experiments and Results sections, and contributed to the paper throughout.

Henry Chen led the VGGT fine-tuning procedure (Condition C), including triplet dataset construction, the AdamW training loop, and mixed-precision training. He produced all figures and visualizations including the recall bar charts, per-scene analysis plots, topometric map overlays, and the 3-D point cloud visualization. He was the primary author of the dataset preparation and runtime benchmarking components, and contributed to the paper throughout.

All three authors jointly drafted the Abstract, Introduction, and Conclusion sections, participated in code review throughout the project, and reviewed the final manuscript.

References

[1] R. Arandjelovic, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, "NetVLAD: CNN architecture for weakly supervised place recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, 2016, pp. 5297–5307.

[2] AI4CE Lab, "NYC-Indoor-VPR dataset," https://huggingface.co/datasets/ai4ce/NYC-Indoor-VPR-Data, 2024.

[3] M. Oquab *et al.*, "DINOv2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, 2024.

[4] H. Wang *et al.*, "VGGT: Visual geometry grounded deep structure from motion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Nashville, TN, 2025.

[5] J. Sivic and A. Zisserman, "Video Google: A text retrieval approach to object matching in videos," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Nice, France, 2003, pp. 1470–1477.

[6] G. Berton, C. Masone, and B. Caputo, "Rethinking visual geo-localization for large-scale applications," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, 2022, pp. 4878–4888.

[7] A. Ali-bey, B. Chaib-draa, and P. Giguere, "MixVPR: Feature mixing for visual place recognition," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Waikoloa, HI, 2023, pp. 2925–2934.

[8] P.-E. Sarlin, C. Cadena, R. Siegwart, and M. Dymczyk, "From coarse to fine: Robust hierarchical localization at large scale," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, 2019, pp. 12716–12725.

[9] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperGlue: Learning feature matching with graph neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, 2020, pp. 4938–4947.

[10] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," *Commun. ACM*, vol. 65, no. 1, pp. 99–106, Jan. 2021.